

a) the closed loop model is

$$M\ddot{q} + C\dot{q} = J^T F + (I - J^T H) \tau_0 \Rightarrow$$

$$\ddot{q} = \bar{M}^{-1} (J^T F + (I - J^T H) \tau_0 - C\dot{q}) \Rightarrow$$

$$\ddot{q} = J \left[\bar{M}^{-1} J^T F + \bar{M}^{-1} (I - J^T H) \tau_0 - \bar{M}^{-1} C \dot{q} \right] + \dot{J} \dot{q} \quad \odot$$

The acceleration given by τ_0 is

$J \bar{M}^{-1} (I - J^T H) \tau_0$ this must be zero so we solve for H

$$\Rightarrow J \bar{M}^{-1} J^T H \tau_0 = J \bar{M}^{-1} \tau_0 \Rightarrow J \bar{M}^{-1} J^T H = J \bar{M}^{-1} \Rightarrow H = (J \bar{M}^{-1} J^T)^{-1} J \bar{M}^{-1} \quad \blacksquare$$

b) Consider again $\odot$, we premultiply by $(J \bar{M}^{-1} J^T)^{-1}$

$$\Rightarrow (J \bar{M}^{-1} J^T)^{-1} \ddot{q} = F \underbrace{(J \bar{M}^{-1} J^T)^{-1} J \bar{M}^{-1} (I - J^T H) \tau_0}_{\text{this term is null, we get}} - (J \bar{M}^{-1} J^T)^{-1} J \bar{M}^{-1} C \dot{q} + (J \bar{M}^{-1} J^T)^{-1} \dot{J} \dot{q}$$

$$\underbrace{(J \bar{M}^{-1} J^T)^{-1} \ddot{q}}_{M_{\tilde{q}}} + \underbrace{(J \bar{M}^{-1} J^T)^{-1} [J \bar{M}^{-1} C \dot{q} - \dot{J} \dot{q}]}_{C_{\tilde{q}}} = 0 \quad \blacksquare$$

Exercise 2

Is trivial that $\omega_1 = \dot{q}_1$ $\omega_2 = \dot{q}_1$ $\omega_3 = \dot{q}_3$ $T_1 = \frac{1}{2} I_1 \dot{q}_1^2$

$$P_{C2} = \begin{cases} (q_2 - \frac{L_2}{2}) C_1 \\ (q_2 - \frac{L_2}{2}) S_1 \end{cases} \quad \dot{P}_{C2} = \begin{cases} \dot{q}_2 C_1 - (q_2 - \frac{L_2}{2}) S_1 \dot{q}_1 \\ \dot{q}_1 S_1 + (q_2 - \frac{L_2}{2}) C_1 \dot{q}_1 \end{cases} \Rightarrow T_2 = \frac{1}{2} I_2 \dot{q}_1^2 + \frac{1}{2} m_2 \left[\dot{q}_2^2 + \dot{q}_1 \left(q_2^2 + \frac{L_2^2}{4} - 2q_2 \frac{L_2}{2} \right) \right]$$

$$P_{C3} = \begin{cases} q_2 C_1 + \frac{L_3}{2} C_3 \\ q_2 S_1 + \frac{L_3}{2} S_3 \end{cases} \quad \dot{P}_{C3} = \begin{cases} \dot{q}_2 C_1 - q_2 S_1 \dot{q}_1 - \frac{L_3}{2} S_3 \dot{q}_3 \\ \dot{q}_1 S_1 + q_2 C_1 \dot{q}_1 + \frac{L_3}{2} C_3 \dot{q}_3 \end{cases}$$

$$T_3 = \frac{1}{2} I_3 \dot{q}_3^2 + \frac{1}{2} m_3 \left[\dot{q}_2^2 + q_2^2 \dot{q}_1^2 + \frac{L_3^2}{4} \dot{q}_3^2 + 2\dot{q}_2 \dot{q}_3 \frac{L_3}{2} \sin(q_1 - q_3) + 2\dot{q}_1 \dot{q}_3 \frac{L_3}{2} q_2 \cos(q_1 - q_3) \right] \Rightarrow$$

$$M = \begin{bmatrix} I_1 + I_2 + m_2 \left(q_2^2 + \frac{L_2^2}{4} - q_2 L_2 \right) + m_3 q_2^2 & 0 & \frac{1}{2} m_3 L_3 q_2 \cos(q_1 - q_3) \\ 0 & m_2 + m_3 & \frac{1}{2} m_3 L_3 \sin(q_1 - q_3) \\ \frac{1}{2} m_3 L_3 q_2 \cos(q_1 - q_3) & \frac{1}{2} m_3 L_3 \sin(q_1 - q_3) & I_3 + \frac{1}{4} L_3^2 \end{bmatrix}$$

Now : derive $q(q) : U = g_0 [m_2 (q_2 - \frac{L_2}{2}) S_1 + m_3 (q_2 S_1 + \frac{L_3}{2} S_3)]$

$$\frac{\partial U}{\partial q_1} = g_0 m_2 (q_2 - \frac{L_2}{2}) C_1 + g_0 m_3 q_2 C_1 = g_0 C_1 (m_2 (q_2 - \frac{L_2}{2}) + m_3 q_2) = g_0 C_1 (e_1 (q_2 - \frac{L_2}{2}) + e_2 q_2)$$

$$\frac{\partial U}{\partial q_2} = g_0 S_1 (m_2 + m_3) = g_0 S_1 (e_1 + e_2)$$

$$\frac{\partial U}{\partial q_3} = \frac{1}{2} g_0 m_3 L_3 S_3 = \frac{1}{2} g_0 L_3 S_3 e_2$$

Now i search the equilibria

$$\begin{cases} g_0 c_1 (e_1 (q_2 - \frac{L_2}{2}) + e_2 q_2) = 0 \\ g_0 s_1 (e_1 + e_2) = 0 \\ \frac{1}{2} g_0 L_3 s_3 e_2 = 0 \end{cases} \begin{cases} e_1 (q_2 - \frac{L_2}{2}) = -e_2 q_2 \Rightarrow q_2 = \frac{e_1 L_2}{2(e_1 + e_2)} \\ s_1 = 0 \Rightarrow q_1 \in \{0, \pi\} \\ s_3 = 0 \Rightarrow q_3 \in \{0, \pi\} \end{cases}$$

A linear parametrization is

$$\begin{pmatrix} g_0 c_1 (q_2 - \frac{L_2}{2}) \\ g_0 s_1 \\ 0 \end{pmatrix} \begin{pmatrix} g_0 c_1 q_2 \\ g_0 s_1 \\ \frac{1}{2} g_0 L_3 s_3 \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = g(q)$$

Now we consider

$$G = \frac{\partial g}{\partial q} = \begin{bmatrix} -g_0 s_1 (e_1 (q_2 - \frac{L_2}{2}) + e_2 q_2) & g_0 c_1 (e_1 + e_2) & 0 \\ g_0 c_1 (e_1 + e_2) & 0 & 0 \\ 0 & 0 & \frac{1}{2} g_0 L_3 s_3 e_2 \end{bmatrix}$$

this matrix is for sure bounded by substituting q_2 with L_2 and $c_i = s_i = 1$

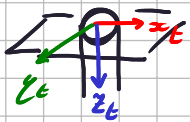
$$\Rightarrow \begin{bmatrix} -g_0 L_2 (e_2 + \frac{1}{2} e_1) & g_0 (e_1 + e_2) & 0 \\ g_0 (e_1 + e_2) & 0 & 0 \\ 0 & 0 & \frac{1}{2} g_0 L_3 e_2 \end{bmatrix}$$

A bound of that matrix is given by the Frobenius norm:

$$\|G\| \leq \alpha = g_0^2 L_2^2 (e_2 + \frac{1}{2} e_1)^2 + 2 g_0^2 (e_1 + e_2)^2 + \frac{1}{4} g_0^2 L_3^2 e_2^2$$

Exercise 3

In this case the sphere can be rotated in all the directions but moved only along z . The natural constraints are $v_x = v_y = s_z = \mu_x = \mu_y = \mu_z = 0$. As artificial constraints we set $\omega_x = \omega_y = \omega_z = s_x = s_y = 0$, $v_z = v(t) \in \mathbb{R}$.



A base of admissible twists and wrenches are

$$\mathcal{B}(V) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \mathcal{B}(F) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

A measurement can be inconsistent when the sensor frame doesn't match with the task frame:

in the control law we have to consider the transformation matrix from the sensor frame to the task frame in order to have a measurement consistent with the geometric model and the task frame.

